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Personalized Outfit Recommendation Workflow

Introduction

Artificial Intelligence is transforming the fashion industry by enabling personalized shopping experiences. Modern fashion platforms use recommendation systems to understand customer preferences and suggest relevant products. However, recommendations based only on purchase history often fail to capture individual style preferences and body characteristics.

To overcome this limitation, modern recommendation systems combine user preferences, body measurements, and visual information to generate more accurate and personalized outfit suggestions.

Project Goal

The primary goal of this project is to design a workflow that:

- Collects user profile information and fashion preferences.
- Analyzes body measurements from uploaded images.
- Extracts meaningful user features.
- Generates personalized outfit recommendations.
- Supports virtual try-on experiences.
- Improves user engagement and satisfaction.

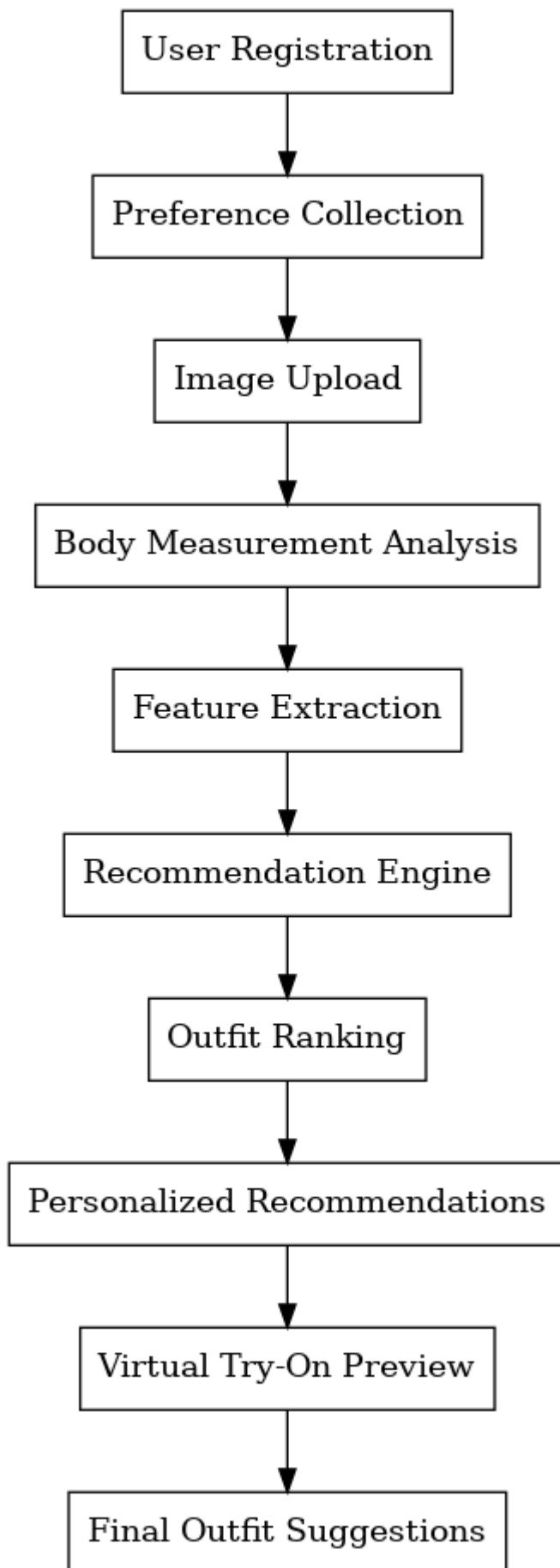
Expected Outcome

A scalable recommendation workflow capable of delivering highly personalized outfit suggestions based on both user preferences and physical characteristics.





Personalized Outfit Recommendation Workflow



Workflow Description

The recommendation workflow consists of multiple stages that work together to generate personalized fashion recommendations.

Step 1: User Registration

The user creates an account and provides basic profile information.

Step 2: Preference Collection

The system collects fashion preferences such as:

- Favorite colors
- Preferred clothing styles
- Brand preferences
- Occasion preferences

Step 3: Image Upload

Users upload a full-body image for analysis.

Step 4: Body Measurement Analysis

Computer vision techniques estimate body measurements and proportions.

Step 5: Recommendation Generation

The recommendation engine uses collected information to identify suitable outfits.

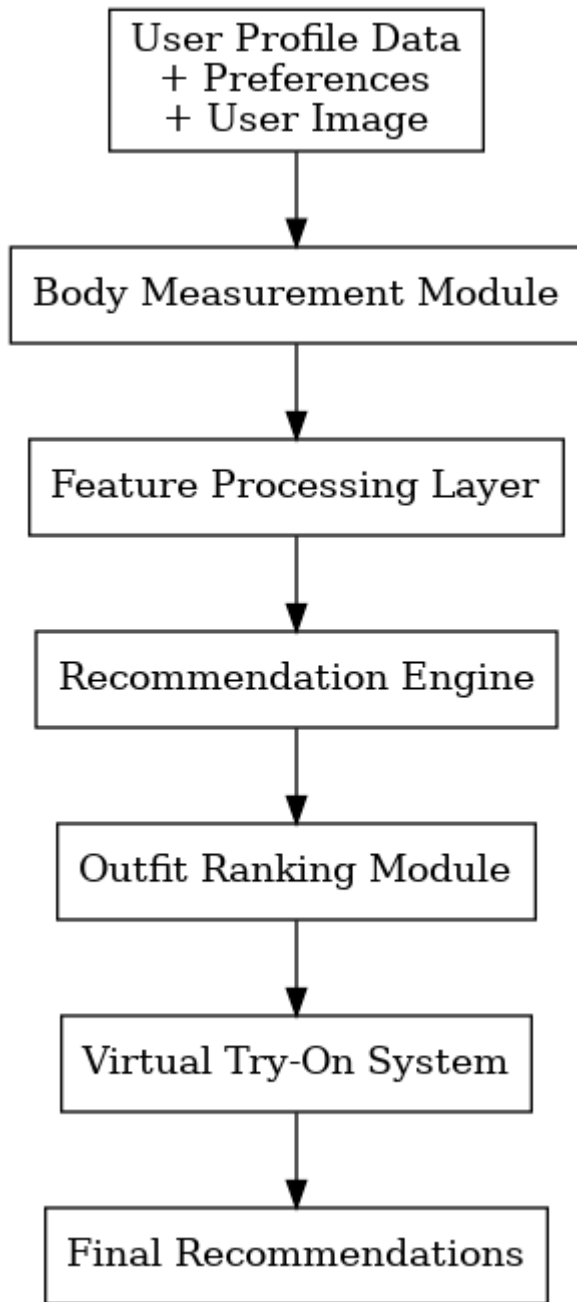
Step 6: Outfit Ranking

Recommendations are ranked according to relevance and compatibility.

Step 7: Virtual Try-On Preview

Users can visualize recommended outfits before making a purchase.

Recommendation System Architecture



Architecture Overview

The architecture is designed to process multiple data sources and generate personalized recommendations efficiently.

Input Layer

The input layer collects:

- User profile information
- Fashion preferences

- Uploaded images

Data Processing Layer

Responsible for:

- Image analysis
- Measurement estimation
- Feature extraction

Recommendation Layer

Processes extracted features and generates outfit recommendations using AI algorithms.

Ranking Layer

Evaluates recommendations using:

- Preference matching
- Style compatibility
- Body measurement compatibility

Output Layer

Displays ranked outfit suggestions and virtual try-on results.

Advantages

- Modular design
- Easy scalability
- Improved recommendation accuracy
- Supports future AI enhancements

Workflow Components in Detail

User Registration

The registration process creates a unique user profile that stores personal and behavioral information.

Collected Data:

- Name
- Age

- Gender
- Contact Information

Importance

Provides the foundation for personalization.

Preference Collection

User fashion preferences help the recommendation engine understand individual tastes.

Examples:

- Casual Wear
- Formal Wear
- Street Fashion
- Traditional Clothing

Importance

Improves recommendation relevance.

Image Upload & Body Analysis

The uploaded image is analyzed to estimate:

- Height
- Shoulder Width
- Body Proportions
- Waist Position

Importance

Improves garment fitting and recommendation quality.

Recommendation Engine & Outfit Ranking



Recommendation Engine

The recommendation engine is the core intelligence component of the workflow.

Content-Based Filtering

Uses product characteristics such as:

- Category
- Color
- Style
- Material

Advantages:

- Personalized
 - Easy to explain
-

Collaborative Filtering

Uses behavior patterns from similar users.

Examples:

- Purchase History
- Wishlist Data
- Product Interactions

Advantages:

- Discovers new products
 - Learns user behavior patterns
-

Hybrid Recommendation

Combines both approaches.

Benefits:

- Higher accuracy
- Better personalization
- Reduced recommendation limitations

Outfit Ranking

Recommendations are ranked based on:

- Preference Match Score
- Body Compatibility Score
- Style Compatibility Score
- Occasion Suitability Score

The highest-scoring outfits are presented to the user

Conclusion

This project focused on designing a complete personalized outfit recommendation workflow for Raritone.

The workflow integrates:

- User profile collection
- Preference analysis
- Body measurement estimation
- AI recommendation techniques
- Outfit ranking mechanisms
- Virtual try-on integration